Reasoning and Learning Across Puzzles and Games

Abstract reasoning in ARC-AGI-2 · Long-horizon learning in ARC-AGI-3

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dots studio

ARC-AGI-2 Abstract reasoning

Infer unfamiliar rules from examples, compose the required transformations, and verify candidate solutions.

dots3-note Preview (Max)

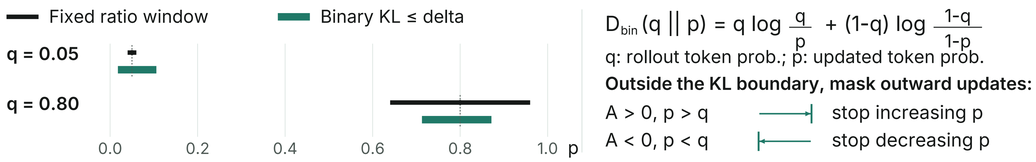
76.8%
ARC Prize Verified [1]

$0.08
Estimated cost / task

Semi-private evaluation · Inference cost at listed token prices [1].

Probability-aware policy updates

We combine selected reasoning demonstrations, supervised initialization and verifiable-reward RL. Binary-KL constraints, adapted from DPPO, regulate token-probability changes during long reasoning [4].



$$D_{bin}(q \,||\, p) = q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p}$$

q: rollout token prob.; p: updated token prob.

Outside the KL boundary, mask outward updates:

A > 0, p > q → stop increasing p

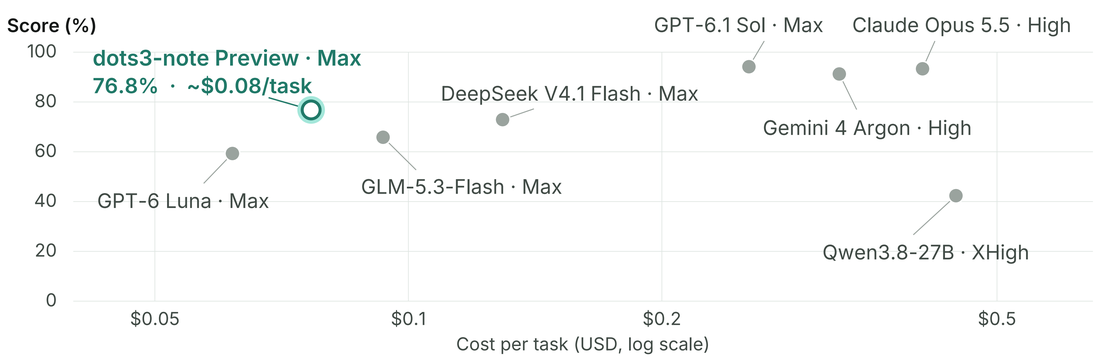
A < 0, p < q → stop decreasing p

Illustration only: epsilon = 0.20, delta = 0.02; not experiment settings. Following DPPO [4]. A: estimated advantage.

Regulate policy updates according to token probability.

Reasoning performance and inference cost

Selected verified results on the semi-private evaluation [1,5]



Highest-scoring verified configuration per selected model. Scores and costs refer to the same configuration; costs follow listed token prices [1,5].

ARC-AGI-3 Long-horizon interaction

Discover unfamiliar game rules through interaction, and plan over long horizons.

dots3-note Preview: ARC-AGI-3 scores [2]

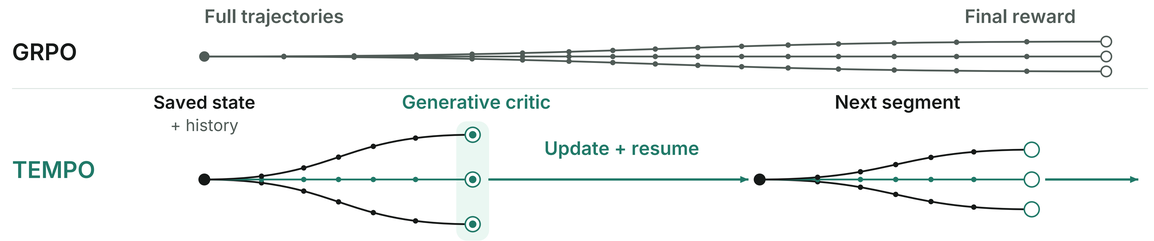
32.1
Claude Code

6.9
Official ARC3 harness

Official public evaluation set · Team self-reported.

TEMPO: Training Before the Task Ends

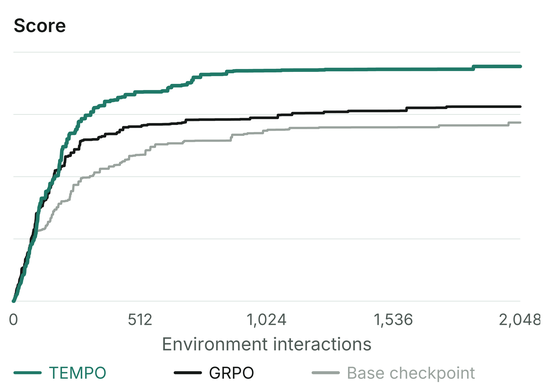
Fork short interaction segments from saved states. A generative critic reasons over the task history to estimate future reward, enabling updates before the episode ends.



Train over short segments. Learn across long horizons.

TEMPO vs. GRPO

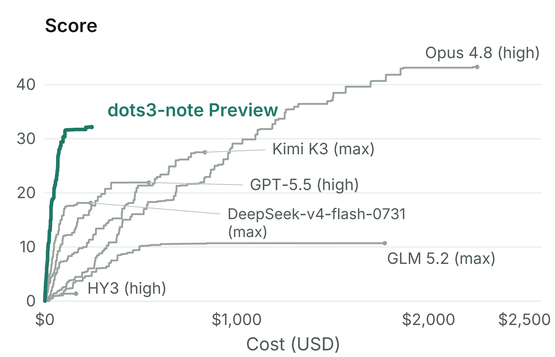
+20.6% relative score gain [2]



Same initialization; 25 games, 2 runs/model/game, up to 2,048 actions [3]. Method comparison, separate from the release-model scores.

Score vs. inference cost

dots3-note Preview and peers [2]



GPT-5.5: Codex; others: Claude Code. Input pricing does not distinguish cached and uncached tokens.

[1] ARC Prize · dots3-note Preview (Max), verified result
https://arcprize.org/results/dots-studio-dots3-note-prev-max

[2] dots studio · dots3-note Preview release report
https://studio.dots.ai/dots/dots3-en.html

[3] dots studio · TEMPO method article
https://studio.dots.ai/dots/tempo-blog.html

[4] Qi et al. · DPPO (2026)
https://arxiv.org/abs/2602.04879

[5] ARC Prize · ARC-AGI-2 leaderboard
https://arcprize.org/leaderboard

Scan for methods and results

TEMPO method

Model report